DERIVED PUSHFORWARD OF A BOUNDED COHERENT COMPLEX WITH PROPER SUPPORT STAYS BOUNDED COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Spectral sequence of a derived functor. Derived Categories, lemma two-ss-complex-functor (cross-chapter, named). R^p f_* H^q(E) => R^{p+q} f_* E.
- Coherence of pushforwards with proper support. Cohomology of Schemes, lemma support-proper-over-base-pushforward (cross-chapter, named). Each R^p f_* H^q(E) is coherent.
- Boundedness from above of higher direct images. Cohomology of Schemes, lemma quasi-coherence-higher-direct-images (cross-chapter, named). Supplies the vanishing of R^p f_* in large degree; alternative via the same-chapter quasi-coherence input.

Source reference. Original reference 01XJ: focused-seeds/geometry/S9-perf2-direct-image-coherent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08DS: focused-seeds/geometry/S9-perf2-direct-image-coherent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 015J: focused-seeds/geometry/S9-perf2-direct-image-coherent/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08D5: sources/stacks-perfect/pinned-original-source.tex, source lines 661–661. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let E be an object of $D^b_{Coh}(\mathcal{O}_X)$ such that the support of $H^i(E)$ is proper over S for all i. Then Rf_*E is an object of $D^b_{Coh}(\mathcal{O}_S)$.*

Proof. Consider the spectral sequence

$$R^p f_* H^q(E) \Rightarrow R^{p+q} f_* E$$

see Derived Categories, Lemma 015J. By assumption and Cohomology of Schemes, Lemma 08DS the sheaves $R^p f_* H^q(E)$ are coherent. Hence $R^{p+q} f_* E$ is coherent, i.e., $Rf_*E \in D_{Coh}(\mathcal{O}_S)$. Boundedness from below is trivial. Boundedness from above follows from Cohomology of Schemes, Lemma 01XJ or from Lemma 08D5. □